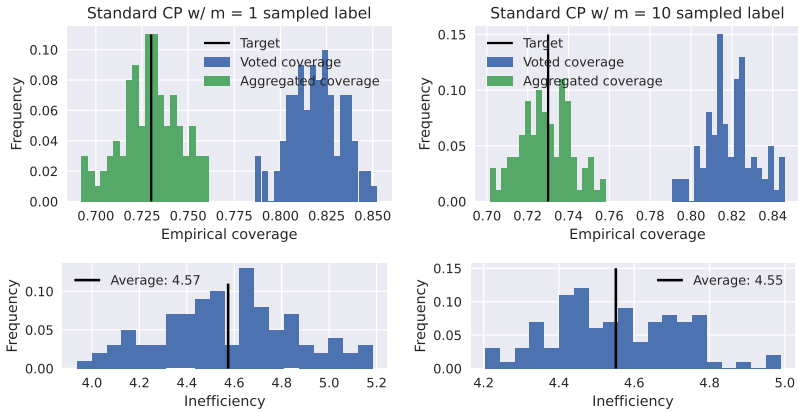

Figure 14: Complementary to Figure 10, we show results for standard CP with $m = 1$ and $m = 10$ sampled labels per example in terms of voted coverage (blue) and aggregated coverage (green) across 100 random calibration/test splits (top) as well as inefficiency (bottom). As for Monte Carlo CP in Algorithm 2, we sample m labels from the plausibilities, but use the quantile from Equation (3) for calibration. For $m = 1$, this coincides with Monte Carlo CP; for $m > 1$, however, the quantile computation differs from Equation (13).

We have

$$\rho_y > \alpha \iff \sum_{j=1}^{m}\sum_{i=1}^{n} \mathbb{I}(S_i^j \leq E(X_{n+1}, y)) > \alpha m(n+1) - m. \tag{29}$$

If $\alpha m(n+1)$ is an integer then $E(X_{n+1}, y)$ needs to be larger or equal than the $\alpha m(n+1)-m+1$ smallest values of $(S_i^j)_{i\in[n],j\in[m]}$. If $\alpha m(n+1)$ is not an integer then $\sum_{j=1}^{m}\sum_{i=1}^{n} \mathbb{I}(S_i^j \leq E(X_{n+1}, y)) \geq \lceil \alpha m(n+1) \rceil - m$, i.e. $E(X, y)$ is larger or equal than the $\lceil \alpha m(n+1) \rceil - m$ smallest values of S_i. However, we have $\lceil \alpha m(n+1) \rceil - 1 = \lfloor \alpha m(n+1) \rfloor$ as $\alpha m(n+1)$ is not an integer. So $\lceil \alpha m(n+1) \rceil - m = \lfloor \alpha m(n+1) \rfloor - m + 1$. So overall we need $E(X_{n+1}, y)$ larger or equal than the $\lfloor \alpha m(n+1) \rfloor - m + 1$ smallest values of $(S_i^j)_{i\in[n],j\in[m]}$, equivalently larger that the quantile of $(S_i^j)_{i\in[n],j\in[m]}$ at level $\frac{\lfloor \alpha m(n+1) \rfloor - m + 1}{mn}$, i.e. $C(X_{n+1}) = \{k \in [K] : E(X_{n+1}, k) \geq \tau\}$ for τ defined in Equation (10).

C Additional results for skin condition classification

Figure 14 shows a baseline experiment using standard CP with sampled labels instead of our Monte Carlo CP from Algorithm 2. Specifically, we sample $m = 1$ and $m = 10$ labels as we would for Monte Carlo CP but apply the standard quantile from Equation (3) for calibration. Specifically, we use $\lfloor \alpha(n+1) \rfloor / n$ (Equation (3)) instead of $\lfloor \alpha m(n+1) \rfloor - m + 1 / mn$ (Equation (13)) which is used for Monte Carlo CP. For $m = 1$, these coincide; for $m > 2$, the quantiles may differ but are still close for large n. Thus, it is not surprising that this approach also obtains (on average) target coverage $1 - \alpha$. However, in contrast to Monte Carlo CP, this approach does not provide any coverage *guarantee* for $m > 1$.

Figure 15 shows the p-values of standard CP (with voted labels, left) and Monte Carlo CP (with sampled labels, right) w.r.t. to the voted labels (top) and 10 labels randomly sampled from the plausibilities (per example, bottom). CP against voted labels results in the corresponding p-values being uniformly distributed. However, the distribution of p-values corresponding to sampled labels is slightly skewed towards 0 (compare to the black line). With Monte Carlo CP, we observe the opposite: the p-values corresponding to voted labels are not entirely uniformly distributed while those corresponding to sampled labels are. This highlights the impact of ambiguity on the corresponding p-values.

Figure 16 presents complementary results to Figure 9 in the main paper. Specifically, on top, we additionally consider ties among the voted labels (i.e., there is no unique $\arg\max$ in the plausibilities λ). In the calibration examples, we break these ties randomly. At test time, however, we can decide to evaluate

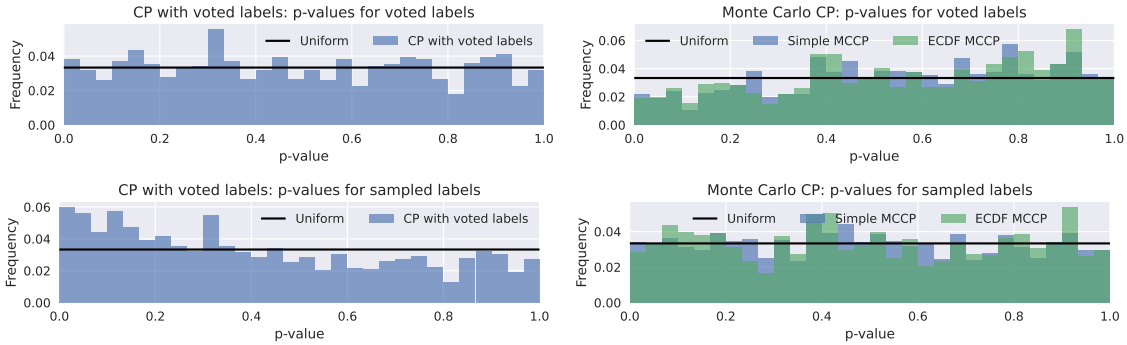

Figure 15: p-value histograms for CP with voted labels (left) and Monte Carlo CP (right). In both cases, we show the p-values for the voted labels (arg max of plausibilities, top) and labels samples from the plausibilities (bottom). Calibrating against voted labels guarantees a uniform distribution of the p-values shown on top, Monte Carlo calibration guarantees the same for the bottom histograms. As a result, compared to the expected uniform distribution (black), the distribution of p-values w.r.t. sampled labels is skewed for CP with voted labels (bottom left) while p-values w.r.t. to voted labels are skewed for Monte Carlo CP (top right).

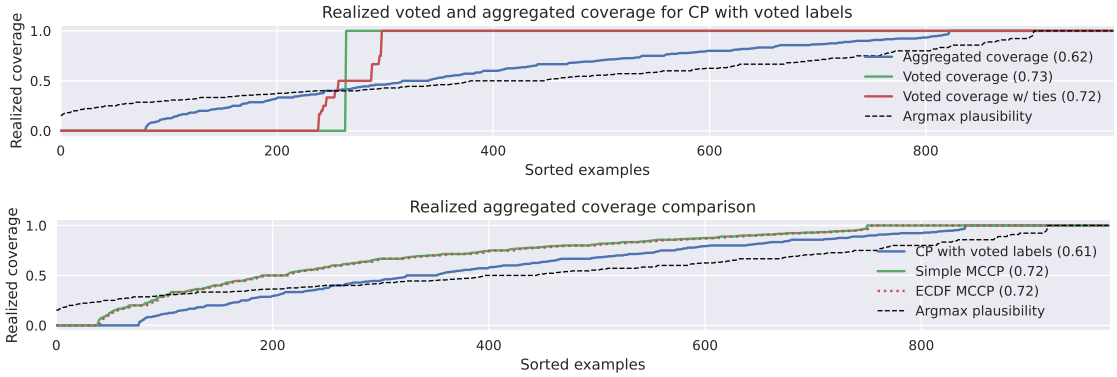

Figure 16: Top: Complementary to Figure 9, we include evaluation against voted labels while considering ties (red, $1/L_i \sum_{j=1}^{L_i} \mathbb{I}[Y_i^j \in C(X_i)]$ with Y_i^j being one of L_i tied labels for case i) which hints towards several ambiguous cases. Looking at aggregated coverage (i.e., $\sum_{y \in [K]} \lambda_{ik} \mathbb{I}[y \in C(X_i)]$) in blue, however, shows that a significant portion of examples are ambiguous and standard evaluation against the voted labels (i.e., $\mathbb{I}[\arg\max_k \lambda_{ik} \in C(X_i)]$) is unreasonable. Bottom: Aggregated coverage plot for Monte Carlo CP procedures compared to CP with voted labels, highlighting that CP with voted labels does not obtain the target aggregated coverage.

coverage proportional to the tied labels. Formally, we assume that $Y_i^1, \ldots, Y_i^{L_i}$ are the tied labels and compute $1/L_i \sum_{j=1}^{L_i} \mathbb{I}[Y_i^j \in C(X_i)]$ instead of the binary indicator $\mathbb{I}[Y_i \in C(X_i)]$ to evaluate coverage. In red, we see that quite a few examples exhibit ties and the predicted prediction sets often cover only a part of the tied labels. This is an early indicator for high ambiguity in the ground truth of this dataset. On the bottom, we additionally plot aggregated coverage for (ECDF) Monte Carlo CP in comparison to CP on voted labels. Again, CP with voted labels significantly under-estimates aggregated coverage. While both Monte Carlo approaches (with and without ECDF correction) look identical in this example, they do not have to be due to randomness in how the calibration set is further split (cf. Algorithm 2). In expectation across splits, however, we found that they coincide. This also highlights that the simple variant empirically achieves aggregated coverage $1 - \alpha$ despite only guaranteeing $1 - 2\alpha$.

Figure 17 reproduces Figure 5 on our skin condition classification case study showing the advantage of using $m > 1$ in Monte Carlo CP to reduce variation in aggregated coverage across calibration/test splits.

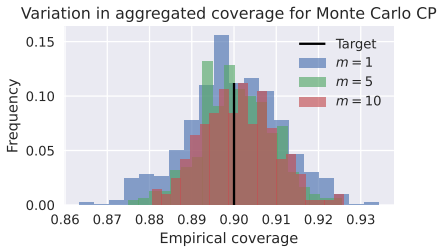

Figure 17: Empirical aggregated coverage for Monte Carlo CP with various m. As on our toy dataset in Figure 5, higher m clearly reduces the variation in coverage across the evaluated 100 calibration/test splits despite using 50% of the 1949 examples for calibration.

Figure 18: Additional qualitative results corresponding to Figure 11.

Figure 18 shows two additional qualitative examples where Monte Carlo CP improves results, i.e., more conditions with significant plausibility (blue) are covered (green) but important conditions are still not covered. This can be addressed using a lower confidence level such as $\alpha = 0.1$ in Figure 20.

Results of our main experiments in dermatology for $\alpha = 0.1$ can be found in Figures 19 to 20.